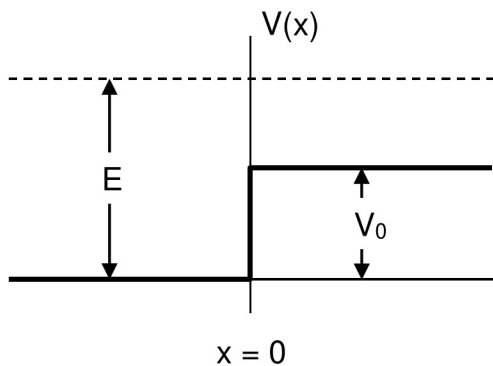


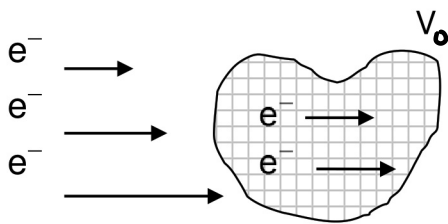
Potenciales seccionalmente constantes

El potencial escalón:

$$V(x) = \begin{cases} 0 & , x < 0 \\ V_0 & , x > 0 \end{cases}$$



Desde el punto vista clásico:



(no hay reflexión, solo hay transmisión)

Ecuación estacionaria de Schrödinger:

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \mu(x) + V(x) \mu(x) = E \mu(x)$$

$$\Rightarrow \frac{d^2}{dx^2} \mu(x) + \frac{2m}{\hbar^2} [E - V(x)] \mu(x) = 0$$

$$\underline{E > V_0}$$

$$\text{Sea } \frac{2mE}{\hbar^2} = k^2, \quad \frac{2m(E - V_0)}{\hbar^2} = q^2$$

$x < 0$:

$$\frac{d^2}{dx^2} \mu(x) + k^2 \mu(x) = 0$$

$$\mu(x) = e^{ikx} + R e^{-ikx}$$

(amplitud de reflexión)

$|R|^2 = R^* R$: coeficiente de reflexión

(Si multiplicamos por $e^{-i\omega t}$, tenemos la variable temporal)

$$x > 0: \frac{d^2 \mu(x)}{dx^2} + q^2 \mu(x) = 0$$

$$\Rightarrow \mu(x) = T e^{iqx} + R e^{-iqx} \quad \{e^{i\omega t}\}$$

(amplitud) (es 0 debido a que no hay reflexión)

$|T|^2$: Coeficiente de transmisión

$x < 0$:

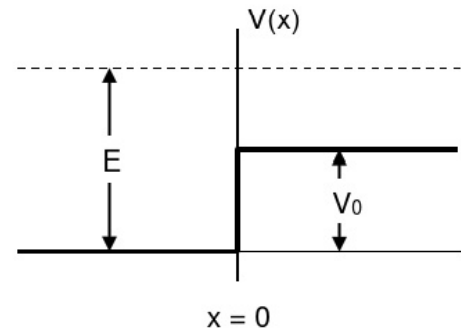
$$j_{iz} = \frac{\hbar}{2im} (\psi^* \nabla \psi - \psi (\nabla \psi^*))$$

Conjugado
hermítico (c.h.)

$$j_{iz} = \frac{\hbar}{2im} [(e^{-ikx} + R^* e^{ikx})(ike^{ikx} - ikR e^{-ikx}) - \text{C.h.}]$$

$$j_{iz} = \frac{\hbar}{2mi} [ik - ikR e^{-2ikx} + ikR^* e^{2ikx} - ikR^* R + ik - ikR^* e^{2ikx} + ikR e^{-2ikx} - ik|R|^2]$$

$$j_{izq} = \frac{\hbar}{2im} [2ik - 2ik|R|^2] = \frac{\hbar k}{m} [1 - |R|^2]$$



$$\mu(x) = e^{ikx} + R e^{-ikx}$$

$\frac{\hbar k}{m}$: flujo incidente

$|R|^2 \frac{\hbar k}{m}$: flujo reflejado por el potencial (cuánticamente hay reflexión)

$$x > 0 : j_{\text{der}} = \frac{\hbar |T|^2 q}{m}$$

Como ρ no depende del tiempo:

$$\left(\nabla \cdot \mathbf{j} + \frac{\partial \rho}{\partial t} = 0 \right)$$

$$\rightarrow \nabla \cdot \mathbf{j} = 0, \quad \frac{\partial j}{\partial x} = 0 \rightarrow j = \text{cte}$$

$$j_{\text{izq}} = j_{\text{der}} \Rightarrow \boxed{\frac{\hbar k}{m} (1 - |R|^2) = \frac{\hbar q}{m} |T|^2}$$

Condiciones de contorno:

1) $\mu_i|_{x=0} = \mu_D|_{x=0}$ * continuidad de la función

2) $\mu_i'|_{x=0} = \mu_D'|_{x=0}$ * continuidad de la derivada de la función

$$\boxed{-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \mu(x) + V(x) \mu(x) = E \mu(x)}$$

$$\mu_i = e^{ikx} + R e^{-ikx}$$

$$\mu_D = T e^{iqx}$$

$$\text{De (1)} : 1 + R = T$$

$$\text{De (2)} : ik - ikR = iqT$$

$$R = \frac{k - q}{k + q}, \quad T = \frac{2k}{k + q}$$

$$\left. \begin{aligned} \frac{\hbar k}{m} |R|^2 &= \frac{\hbar k}{m} \left(\frac{k-q}{k+q} \right)^2 \\ \frac{\hbar q}{m} |T|^2 &= \frac{\hbar k}{m} \frac{4kq}{(k+q)^2} \end{aligned} \right\} \begin{array}{l} \text{(Estos resultados} \\ \text{verifican la} \\ \text{continuidad de la} \\ \text{densidad de} \\ \text{corriente de} \\ \text{probabilidad.)} \end{array}$$

$$\begin{aligned} \frac{\hbar k}{m} (1 - |R|^2) &= \frac{\hbar k}{m} \frac{(k+q)^2 - (k-q)^2}{(k+q)^2} \\ &= \frac{\hbar k}{m} \frac{4kq}{(k+q)^2} = \frac{\hbar q}{m} |T|^2 \end{aligned}$$

$$|R|^2 = \left| \frac{j_R}{j_i} \right|, \quad |T|^2 = \left| \frac{j_T}{j_i} \right|$$

Definiciones
coeficientes de reflexión y transmisión

$$|R|^2 = \left(\frac{k-q}{k+q} \right)^2, \quad k^2 = \frac{2mE}{\hbar^2}$$

$$|T|^2 = \frac{4k^2}{(k+q)^2}, \quad q^2 = \frac{2m(E - V_0)}{\hbar^2}$$

* Hay onda reflejada

* Para $E \gg V_0 \rightarrow k \approx q \rightarrow |R|^2 \rightarrow 0$

$E < V_0$:

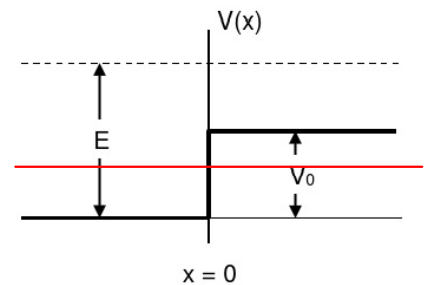
$$\frac{d^2 \mu}{dx^2} + \frac{2m}{\hbar^2} [E - V_0] \mu = 0, \text{ sea } q = \underbrace{\frac{2m}{\hbar^2} [V_0 - E]}_{\text{(positivo)}}$$

$$x > 0: \quad \frac{d^2 \mu_D}{dx^2} - q \mu_D = 0$$

$$\begin{aligned} \rightarrow \mu_D &= A e^{qx} + B e^{-qx} \\ \mu_D &= B e^{-qx} \end{aligned}$$

$$x < 0: \quad \mu_i = e^{ikx} + R e^{-ikx}$$

$$\mu_i|_{x=0} = \mu_D|_{x=0} \rightarrow 1 + R = B \quad (1)$$



$$\mu_1 \Big|_{x=0} = \mu_D \Big|_{x=0} \rightarrow ik - ikR = -Bq \quad (2)$$

$$1 + R = B \quad (1)$$

De (1) y (2): $B = ?$ $R = ?$

$$ik - ik(B - 1) = -Bq$$

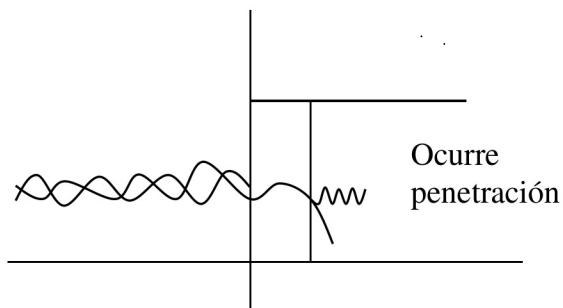
$$(ik - q)B = 2ik \rightarrow$$

$$B = \frac{2ik}{ik - q}$$

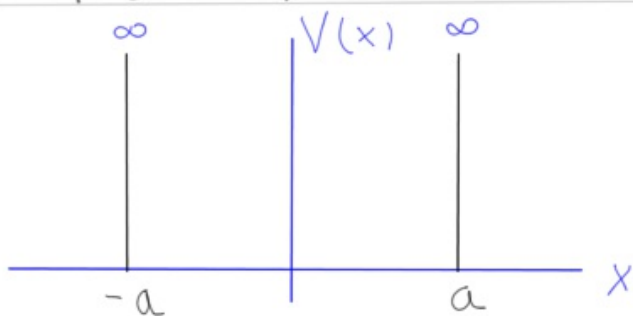
$$R = B - 1 = \frac{2ik - ik + q}{ik - q} = \frac{ik + q}{ik - q}$$

$$|B|^2 = \frac{2ik}{(ik - q)} \frac{-2ik}{(-ik - q)} = \frac{4k^2}{q^2 + k^2}$$

$$|R|^2 = \frac{-ik + q}{-ik - q} \frac{ik + q}{ik - q} = \frac{q^2 + k^2}{q^2 + k^2} = 1 \quad (\text{No hay transmisión})$$



El pozo de potencial infinito



$$V(x) = \begin{cases} 0 & |x| < a \\ \infty & x < -a, x > a \end{cases}$$

$$a = \frac{L}{2}$$

$$\psi(-a) = \psi(a) = 0$$

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi(x)}{dx^2} = E \psi(x) \rightarrow \psi(x) = A \sin kx + B \cos kx; \quad k = \left(\frac{2mE}{\hbar^2} \right)^{1/2}$$

$$\rightarrow A \sin ka = 0 \quad \wedge \quad B \cos ka = 0$$

$$-A \sin ka + B \cos ka = 0$$

$$A \sin ka + B \cos ka = 0$$

$$1) \quad A = 0 \quad \wedge \quad \cos ka = 0 \rightarrow k_n = \frac{n\pi}{2a} = \frac{n\pi}{L}, \quad n = 1, 3, 5, \dots$$

$$\psi_n(x) = \frac{1}{\sqrt{a}} \cos \frac{n\pi x}{2a}, \quad n = 1, 3, 5, \dots$$

$$2) \quad B = 0 \quad \wedge \quad \sin ka = 0 \rightarrow k_n = \frac{n\pi}{2a}, \quad n = 2, 4, 6, \dots$$

$$\psi_n(x) = \frac{1}{\sqrt{a}} \sin \frac{n\pi x}{2a}, \quad n = 2, 4, 6, \dots$$

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2}{8m} \frac{\pi^2 n^2}{a^2} = \frac{\hbar^2}{2m} \frac{\pi^2 n^2}{L^2}, \quad n = 1, 2, 3, \dots$$

$$E_1 = \frac{\hbar^2 \pi^2}{8ma^2}$$

$$\psi(x, t) = \sum_{n=1}^{\infty} c_n \psi_n(x) e^{-\frac{i}{\hbar} E_n t}$$

$$c_n = \int_{-a}^a \psi_n^*(x) \psi(x, t=0) dx$$

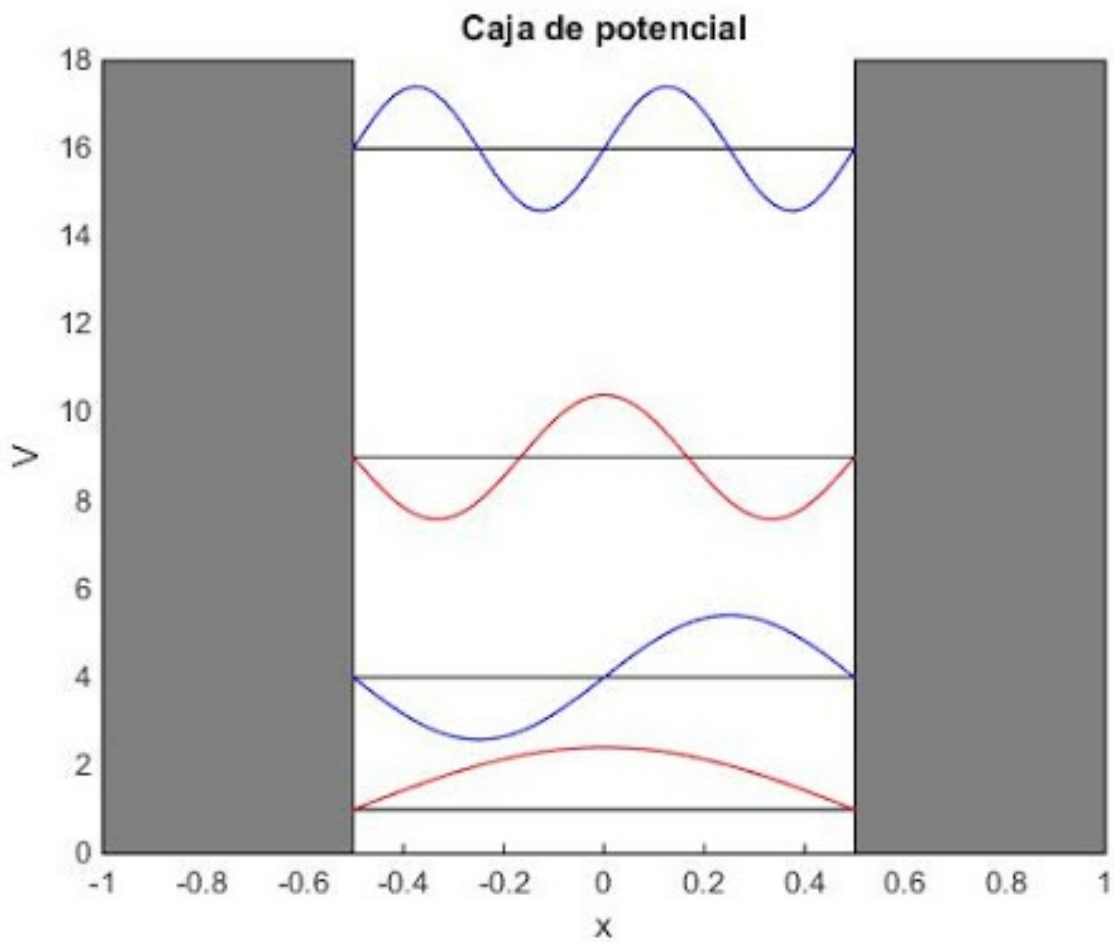
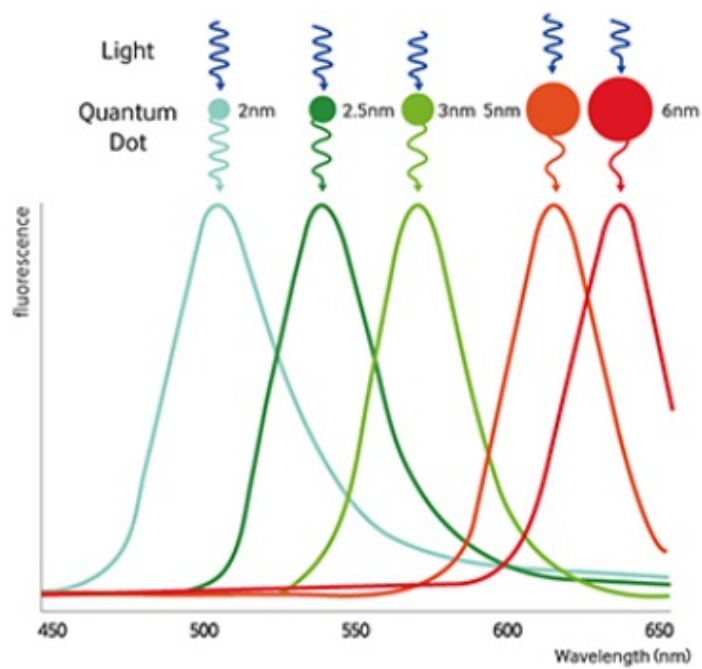


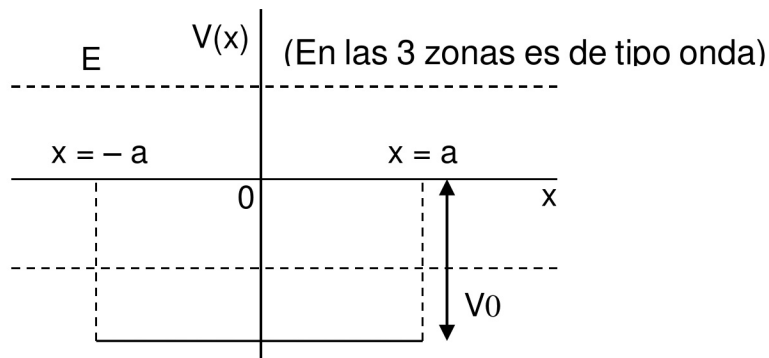
Fig. Niveles de energía en el pozo de potencial infinito

Color of Light Depends On Size of Quantum Dot



El pozo de potencial

$$V(x) = \begin{cases} 0 & x < -a \\ -V_0 & -a < x < a \\ 0 & x > a \end{cases}$$



$$\frac{d^2\mu(x)}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\mu(x) = 0$$

$$\mu(x) = \begin{cases} e^{ikx} + Re^{-ikx} & x < -a \\ Ae^{iqx} + Be^{-iqx} & -a < x < a \\ Te^{ikx} & x > a \end{cases} \quad (E > 0)$$

$$k^2 = \frac{2mE}{\hbar^2}; \quad q^2 = \frac{2m}{\hbar^2}(E + V_0)$$

- $\frac{\hbar k}{m}|R|^2$: flujo reflejado
- $\frac{\hbar k}{m}|T|^2$: flujo transmitido

Conservación de flujo:

$$\frac{\hbar k}{m}(1 - |R|^2) = \frac{\hbar q}{m}(|A|^2 - |B|^2) = \frac{\hbar k}{m}|T|^2$$

Por las condiciones de contorno:

- En $x = -a$:

$$e^{-ika} + Re^{ika} = Ae^{-iqa} + Be^{iqa}$$

$$ike^{-ika} - ikRe^{ika} = iqAe^{-iqa} - iqBe^{iqa}$$

• En $x = a$:

$$Ae^{iqa} + Be^{-iqa} = Te^{ika}$$

$$iqAe^{iqa} - iqBe^{-iqa} = ikTe^{ika}$$

• En $x = -a$:

$$e^{-ika} + Re^{ika} = Ae^{-iqa} + Be^{iqa}$$

$$ike^{-ika} - ikRe^{ika} = iqAe^{-iqa} - iqBe^{iqa}$$

(Ver Gassiorowicz)

$$R = ie^{-2ika} \frac{(q^2 - k^2) \sin 2qa}{2kq \cos 2qa - i(q^2 + k^2) \sin 2qa}$$

$$T = e^{-2ika} \frac{2kq}{2kq \cos 2qa - i(q^2 + k^2) \sin 2qa}$$

Si $E \gg V_0$: $\rightarrow q \approx k \Rightarrow R \rightarrow 0$

$$k^2 = \frac{2mE}{\hbar^2}; \quad q^2 = \frac{2m}{\hbar^2}(E + V_0)$$

Si $E \rightarrow 0$: $\Rightarrow k \rightarrow 0 \Rightarrow T \rightarrow 0$

Si $\sin 2qa = 0 \rightarrow R = 0$

$$2qa = n\pi \rightarrow q = \frac{n\pi}{2a}$$

Esto aparece en la dispersión de electrones por átomos de gases nobles (neón, argón)

$$q^2 = \frac{2m}{\hbar^2}(E + V_0)$$

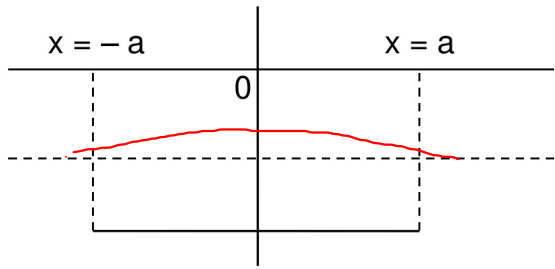
$$E = -V_0 + \frac{\hbar^2}{2m} \left(\frac{n\pi}{2a} \right)^2 = -V_0 + \frac{\hbar^2 \pi^2}{8ma^2} n^2$$

Para $E < 0$:

$$\mu(x) = \begin{cases} C_1 e^{kx} & x < -a \\ A \cos qx + B \sin qx & -a < x < a \\ C_2 e^{-kx} & x > a \end{cases}$$

$$\frac{d^2 \mu(x)}{dx^2} + \frac{2m}{\hbar^2} [E - V(x)] \mu(x) = 0$$

En el primer y tercer término se colocan estos resultados ya que se quiere tener funciones de cuadrado integrable.



$$k^2 = -\frac{2mE}{\hbar^2}$$

$$q^2 = \frac{2m}{\hbar^2}(V_0 + E)$$

$$k^2 = \frac{2m|E|}{\hbar^2}$$

$$q^2 = \frac{2m}{\hbar^2}(V_0 - |E|)$$

Condiciones de contorno:

$$\begin{cases} C_1 e^{-ka} = A \cos qa - B \sin qa \\ k C_1 e^{-ka} = q(A \sin qa + B \cos qa) \end{cases}$$

$$\begin{cases} C_2 e^{-ka} = A \cos qa + B \sin qa \\ -k C_2 e^{-ka} = -q(A \sin qa - B \cos qa) \end{cases}$$

$$\mu(x) = \begin{cases} C_1 e^{kx} & x < -a \\ A \cos qx + B \sin qx & -a < x < a \\ C_2 e^{-kx} & x > a \end{cases}$$

* Estado estacionario

* Estado ligado (condición de normalización) → Discretización de la energía.
Relacionado a un confinamiento – ligadura.

$$\begin{aligned} k &= q \frac{A \sin qa - B \cos qa}{A \cos qa + B \sin qa} \\ &= q \frac{A \sin qa + B \cos qa}{A \cos qa - B \sin qa} \end{aligned}$$

Soluciones pares B = 0:

$$k^2 = \frac{2m|E|}{\hbar^2} \quad q^2 = \frac{2m}{\hbar^2}(V_0 - |E|)$$

$$k = q \tan qa \quad (*)$$

$$\text{Sea } \lambda = \frac{2mV_0 a^2}{\hbar^2} \text{ e } y = qa$$

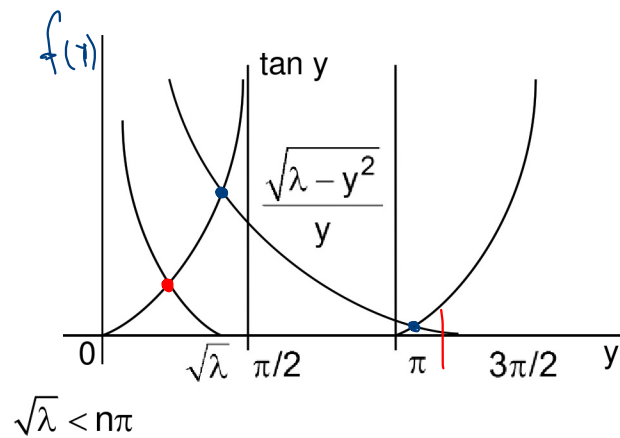
$$\rightarrow q^2 = \frac{\lambda}{a^2} - k^2$$

$$k^2 = \frac{\lambda}{a^2} - q^2 = \frac{\lambda - y^2}{a^2}$$

$$\frac{\sqrt{\lambda - y^2}}{a} = q \tan y \quad \text{ó} \quad \frac{\sqrt{\lambda - y^2}}{y} = \tan y$$

$$\Rightarrow \boxed{\frac{\sqrt{\lambda - y^2}}{y} = \tan y}$$

$$k^2 = \frac{2m|E|}{\hbar^2} \quad q^2 = \frac{2m}{\hbar^2}(V_0 - |E|)$$

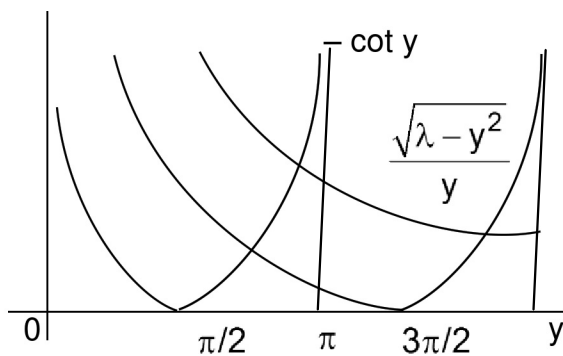


$$\lambda = \frac{2mV_0a^2}{\hbar^2}$$

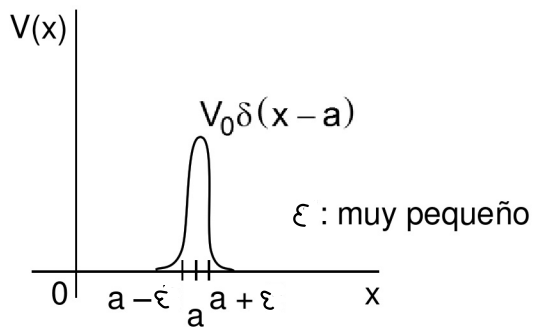
Ejm: $\sqrt{\lambda} < \pi$ (corresponde a una única solución)(un solo corte)

Soluciones impares $A = 0$:

$$\frac{\sqrt{\lambda - y^2}}{y} = -\cot y$$

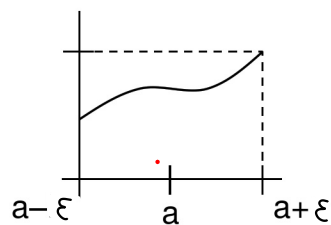


Continuidad de la derivada:



$$\left(\frac{d\mu}{dx} \right)_{a+\epsilon} - \left(\frac{d\mu}{dx} \right)_{a-\epsilon} = \int_{a-\epsilon}^{a+\epsilon} dx \frac{d}{dx} \frac{d\mu}{dx}$$

$$\int_{a-\epsilon}^{a+\epsilon} \frac{d^2\mu}{dx^2} d\mu = \frac{2m}{\hbar^2} \int_{a-\epsilon}^{a+\epsilon} dx \underbrace{(V(x) - E)}_{\text{}} \mu(x)$$



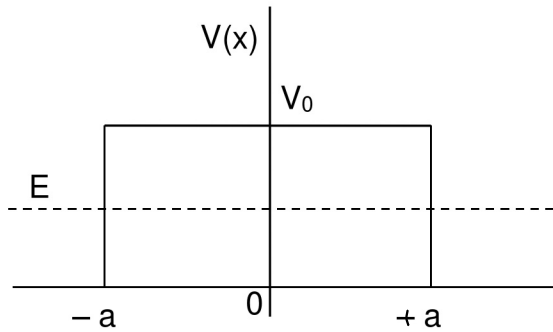
Por el Teorema del
V.M. se garantiza la
continuidad de $\frac{d\mu}{dx}$

En el caso de $\delta(x-a) V_0$

$$\left(\frac{d\mu}{dx} \right)_{a+\epsilon} - \left(\frac{d\mu}{dx} \right)_{a-\epsilon} = \frac{2m}{\hbar^2} \int_{a-\epsilon}^{a+\epsilon} V_0 \delta(x-a) \mu(x) dx$$

$$= \frac{2mV_0}{\hbar^2} \mu(a)$$

El potencial tipo barrera:



Efecto túnel cuántico

$$V_{(x)} = \begin{cases} 0 & x < -a \\ V_0 & -a < x < a \\ 0 & x > a \end{cases}$$

Metal vacío Metal (Por efecto túnel hay corriente entre los 2 metales.)

$$\frac{d^2\mu(x)}{dx^2} + \frac{2m}{\hbar^2}(E - V_0)\mu(x) = 0$$

$$q^2 = \frac{2m}{\hbar^2}(V_0 - E)$$

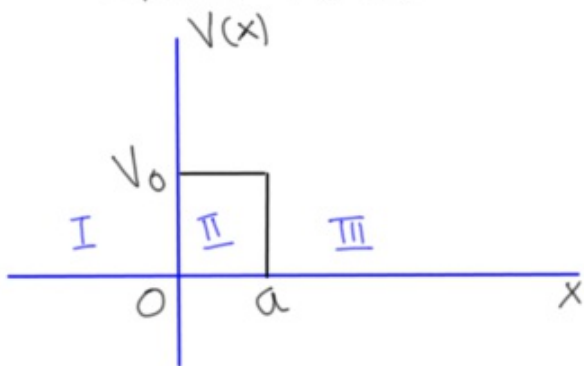
$$\frac{d^2\mu(x)}{dx^2} - \frac{2m}{\hbar^2}(V_0 - E)\mu(x) = 0$$

$$\mu(x) = \begin{cases} e^{ikx} + Re^{-ikx} & x < -a \\ Ae^{qx} + Be^{-qx} & x \in [-a, a] \\ Te^{ikx} & x > a \end{cases}$$

$$k = \sqrt{\frac{2m}{\hbar^2}E}, \quad q = \sqrt{\frac{2m}{\hbar^2}(V_0 - E)}$$

Se obtiene $T = \frac{1}{1 + \frac{1}{4} \left(\frac{V_0^2}{E(V_0 - E)} \right) \sinh^2 qa}$

Efecto Túnel



$$V = \begin{cases} 0 & x < 0 \\ V_0 & 0 \leq x \leq a \\ 0 & x > a \end{cases}$$

$$\frac{\partial^2 \psi_I}{\partial x^2} + \frac{2m}{\hbar^2} E \psi_I = 0, \quad x < 0$$

$$\frac{\partial^2 \psi_{II}}{\partial x^2} + \frac{2m}{\hbar^2} (E - V_0) \psi_{II} = 0 \quad 0 \leq x \leq a$$

$$\frac{\partial^2 \psi_{III}}{\partial x^2} + \frac{2m}{\hbar^2} E \psi_{III} = 0 \quad x > a$$

$$E < V_0$$

$$\psi_I(x) = A e^{ikx} + B e^{-ikx} \quad x < 0$$

$$\psi_{II}(x) = C e^{\beta x} + D e^{-\beta x} \quad 0 \leq x \leq a$$

$$\psi_{III}(x) = F e^{ikx} \quad x > a$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$\beta = \sqrt{\frac{2m}{\hbar^2} (V_0 - E)}$$

En $x=0$:

$$\psi_I(0) = \psi_{II}(0) \rightarrow A + B = C + D \quad \dots (1)$$

$$\left. \frac{d\psi_I}{dx} \right|_{x=0} = \left. \frac{d\psi_{II}}{dx} \right|_{x=0} \rightarrow i k A - i k B = \beta C - \beta D \quad \dots (2)$$

En $x=a$:

$$\psi_{II}(a) = \psi_{III}(a) \rightarrow C e^{\beta a} + D e^{-\beta a} = F e^{ik a} \quad \dots (3)$$

$$\left. \frac{d\psi_{II}}{dx} \right|_{x=a} = \left. \frac{d\psi_{III}}{dx} \right|_{x=a} \rightarrow \beta C e^{\beta a} - \beta D e^{-\beta a} = i k F e^{ik a} \quad \dots (4)$$

$$\frac{(3)}{A} + \frac{(4)}{\beta A} : \quad \frac{C}{A} = \frac{1}{2} \frac{E}{A} e^{-\beta a} \left[1 + \frac{i k}{\beta} \right] e^{ik a}$$

$$\frac{(3)}{A} - \frac{(4)}{\beta A} : \quad \frac{D}{A} = \frac{1}{2} \frac{E}{A} e^{\beta a} \left[1 - \frac{i k}{\beta} \right] e^{ik a}$$

Sustituyendo en $\frac{(1)}{A}$ y en $\frac{(2)}{A}$

$$1 + \frac{B}{A} = \frac{F}{A} \left[\cosh \beta a - i \frac{k}{\beta} \sinh \beta a \right] e^{i k a}$$

$$1 - \frac{B}{A} = \frac{F}{A} \left[\cosh \beta a - \frac{\beta}{i k} \sinh \beta a \right] e^{i k a}$$

Sumando

$$\frac{A}{F} = \left[\cosh \beta a + \frac{i}{2} \left(\frac{\beta}{k} - \frac{k}{\beta} \right) \sinh \beta a \right] e^{i k a}$$

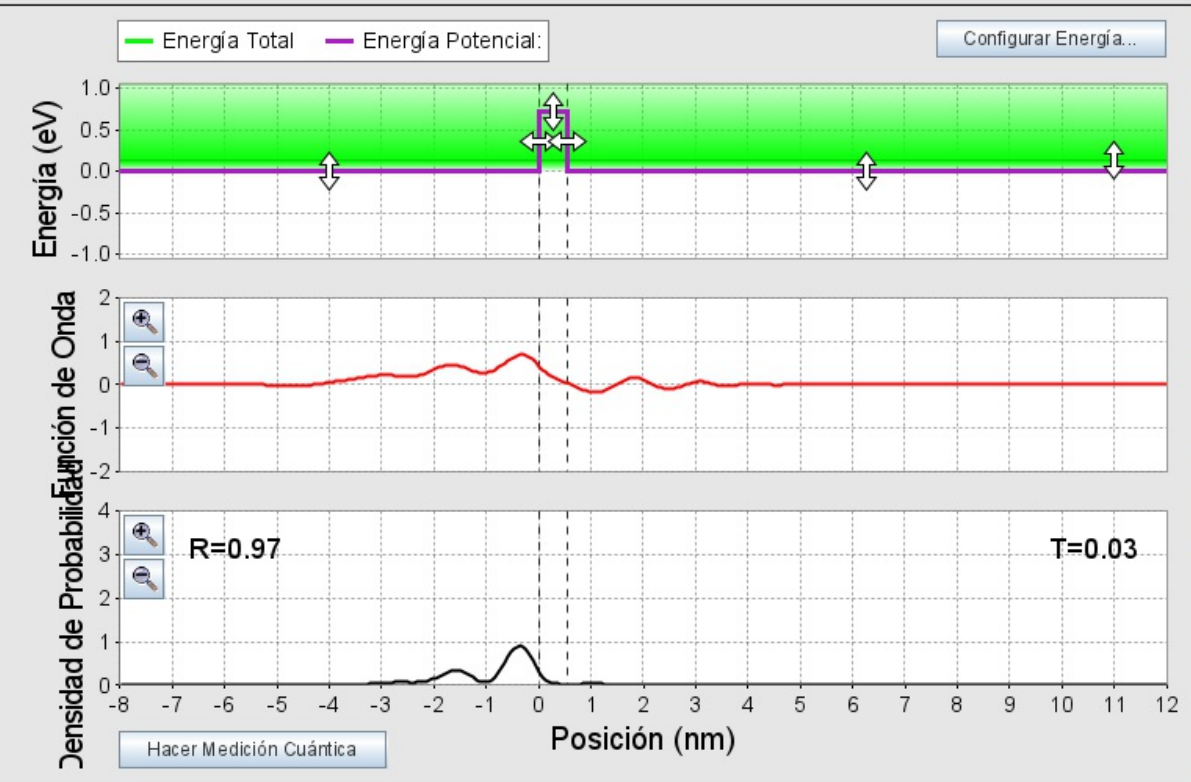
$$T = \left| \frac{F}{A} \right|^2 \rightarrow \frac{1}{T} = 1 + \frac{1}{4} \left(\frac{\beta^2 + k^2}{k\beta} \right)^2 \sinh^2 \beta a$$

Com J

$$K = \sqrt{\frac{2mE}{\hbar^2}} \quad \wedge \quad \beta = \sqrt{\frac{2m}{\hbar^2} (V_0 - E)}$$

$$\rightarrow \left(\frac{\beta^2 + k^2}{k\beta} \right)^2 = \frac{V_0^2}{E(V_0 - E)}$$

$$\frac{1}{T} = 1 + \frac{1}{4} \left(\frac{V_0^2}{E(V_0 - E)} \right) \sinh^2 \beta a$$



barrera/pared

☐ Mostrar Valores de Energía

☒ Mostrar probabilidades de reflexión y transmisión

Vista del función de la onda del electrón:

☒ Parte Real

☐ Parte Imaginaria

☐ Magnitud

☐ Fase 0 2π

Dirección de la onda entrante:

Forma del función de la onda del electrón:

☒ Paquete de Onda

☐ Onda plana

Propiedades del Paquete de Onda

Anchura inicial σ : 0.5 nm

Posición Inicial -2.0 nm

Reset All

Help!

7.8 fs